# Sajeevan Veeriah

## Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer | MES/ERP Integration, Embedded Design & Control | Applied R&D

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## PROFILE

Mechatronics engineer (Honours, Distinction) who integrated PLC/SCADA and MES systems in GMP manufacturing at JAG Process Solutions and validated vehicle software and ADAS at Ford. Since June 2026, client work through Voltwyn has spanned vehicle electronics, firmware and full-stack software. Also brings industrial networking, commissioning, embedded systems, PCB design and ROS 2 robotics skills.

**Core strengths:** Industrial IT/OT and MES/ERP integration | Industrial automation and controls | Robotics and autonomy | Embedded systems and PCB design | Vehicle test and validation | Software, AI/ML and data | Commissioning, quality and compliance

## TECHNICAL SKILLS

**Industrial IT/OT and systems:** Manufacturing execution systems (MES), ERP integration, SCADA and historian platforms, OEE, downtime and production reporting, traceability and master data, industrial networking (Ethernet, switches, servers), OT cyber security, backup and disaster recovery, system testing and validation, vendor coordination.

**Automation and controls:** PLC logic and integration, Siemens TIA Portal, HMI/SCADA (iFIX, PVI+), drives, Modbus, PROFINET, FAT/SAT, commissioning, GMP, ITPs, MDRs and AS/NZS compliance.

**Robotics and embedded:** ROS 2, Nav2, SLAM, EKF sensor fusion, Gazebo, RViz, C/C++, ESP32/ESP32-S3, STM32, FreeRTOS, CAN/CAN FD, UART, I2C, SPI and BLE.

**Software, AI and data:** Python, MATLAB/Simulink, TypeScript, React, Rust (Tauri), SQL/SQLite, Git, CI/CD, OpenCV, scikit-learn and local language-model retrieval, memory and evaluation workflows.

**Hardware, IoT and test:** Altium, KiCad, SolidWorks, Fusion 360, PCB and enclosure design, 3D printing, Linux, LoRaWAN, ChirpStack, MQTT, InfluxDB, Grafana, CANoe, CANalyzer and calibrated data acquisition.

## PROFESSIONAL EXPERIENCE

### Automation and Controls Engineer
Jan 2026 - Jun 2026
**JAG Process Solutions**

- Integrated PLC logic, HMI/SCADA, field devices, drives and manufacturing execution system (MES) data for pharmaceutical, biotech and food plants.
- Migrated validated iFIX content to PVI+ and verified functionality; supported factory and site acceptance testing (FAT/SAT), commissioning and GMP handover evidence.
- Isolated MiR autonomous mobile robot faults across control, network and production interfaces to identify the responsible system layer.

### Product Development Test Engineer (Contract)
Oct 2025 - Jan 2026
**Ford Motor Company via Invenio**

- Tested vehicle-software integration and advanced driver assistance systems (ADAS) through feature-vehicle, breadboard and regression testing.
- Instrumented vehicles and analysed CAN/CAN FD data in CANoe and CANalyzer to isolate faults.

### Technical Officer & Quality Assurance
Jul 2024 - Aug 2025
**ABMARC**

- Conducted emissions, EV/PHEV range and vehicle-system tests using calibrated instrumentation and data acquisition (DAQ).
- Maintained traceable test, safety and quality records for certification review.

### Consultant Engineer, IoT and Projects Administrator
Feb 2024 - Aug 2024
**DuxTel**

- Built ESP32 and LoRaWAN AU915 telemetry for agricultural equipment, capturing CAN and GPS data through ChirpStack, MQTT, InfluxDB and Grafana.
- Integrated custom PCBs, MikroTik connectivity and Linux services for provisioning, field trials and handover.

## SV TECH SOLUTIONS: CLIENT PROJECTS (JUN 2026 - PRESENT)

**Vehicle controller and display system** | Electronics and firmware
Designed a four-layer ESP32-S3 vehicle controller, serial-adapter PCB, display firmware, browser configuration and vehicle-position sensing. Prepared fabrication files; assembled-system and field validation remain pending.

**Pricing and inventory software** | Full-stack software
Developed pricing and inventory software as sole engineer (React/TypeScript, Tauri/Rust, SQLite). Built operator-reviewed imports into a job-management platform with change, exception, rollback and audit reports.

## MANUFACTURING AND QUALITY EXPERIENCE

**Production Line Worker / Supervisor / Commissioning Support; Cellar Hand** Aug 2022 - Feb 2024
**IDL Australia**

- Supervised production output, changeovers and handovers; maintained quality and traceability and supported canning-line commissioning to Sep 2023.
- From Sep 2023, supported cellar sanitation, sampling and production records.

**Production Line Operator and Commissioning Support** Sep 2021 - Dec 2021
**Carbon Revolution**

- Operated carbon-fibre wheel equipment and maintained quality records; supported equipment trials, setup and first-level fault recovery.

**Undergraduate Quality Engineer** Mar 2018 - Mar 2020
**Thornton Engineering Australia**

- Supported drawing review, ITPs, MDRs, material traceability and AS/NZS compliance in structural-steel fabrication.

## SELECTED ENGINEERING PROJECTS

Full portfolio: sv.sajeevanveeriah.workers.dev/work

**Autonomous navigation rover** | ROS 2 autonomy
Built a differential-drive platform with LiDAR and IMU. Integrated SLAM, EKF localisation and Nav2 in C++/Python; verified navigation and recovery in Gazebo and RViz. Autonomy validated in simulation.

**Ataxia assessment device** | Embedded mechatronics
Designed an ESP32 prototype, Altium PCB and enclosure with four Hall-effect channels sampled at 100 Hz and Bluetooth reporting. Tested accuracy, reversals, drift and temperature behaviour in MATLAB.

**SPC-001 waterless solar-panel cleaner** | Robotics
Developed mechanical CAD, electronics, supervisory control and coverage planning. Wrote an independent research paper on contact mechanics, traction and energy that distinguishes analysis from physical performance.

**VeerAI** | Local AI system
Built a locally deployed language-model system with document ingestion, retrieval, memory, tools and evaluation.

## EDUCATION

**Bachelor of Mechatronics Engineering (Honours), Distinction** 2025
**Deakin University**

**Higher National Diploma in Mechatronics, Robotics and Automation Engineering, Distinction** 2016
**Cardiff Metropolitan University**

## MEMBERSHIP AND LANGUAGES

Member, Engineers Australia | English, Tamil and Sinhala | Community leadership and engineering mentoring.